

TWO ABSOLUTELY BOUNDED DETERMINANTAL RATIOS

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ABSTRACT. Bounded ratios of products of minors of positive definite matrices have a long history, starting with Hadamard's inequality in 1893. It states that for every positive semidefinite matrix A

$$\det A \leq A_{11} \cdots A_{nn}.$$

This inequality was subsequently generalized by Fisher and then further by Kotljanskii. The latter states that for every positive semidefinite matrix A and any index sets $\alpha_1, \alpha_2 \subseteq \{1, \dots, n\}$ one has

$$\det A[\alpha_1 \cup \alpha_2] \det A[\alpha_1 \cap \alpha_2] \leq \det A[\alpha_1] \det A[\alpha_2],$$

where $A[\alpha]$ denotes the principal submatrix determined by the indexes in α .

In a manuscript published only on the arXiv in 2008, Hall and Johnson made three conjectures about ratios of products of principal minors of 4×4 positive definite matrices, denoted by R_i , $i = 1, 2, 3$, see (2) and (3). They hypothesized that the supremum of R_1 was $27/16$, while the supremum of the other two ratios was 1. Such ratios are called *absolutely bounded*. The conjecture for R_1 was affirmed in [17] and it is the only known bounded determinantal ratio with supremum bigger than one. The goal of this paper is to affirm the conjecture for R_2 and R_3 .

It is known that the upper bound for the ratios R_i , $i = 1, 2, 3$, does not follow from repeated applications of Kotljanskii's inequality. In addition, Hall and Johnson showed that R_i is bounded above by 4, for $i = 1, 2, 3$.

1. INTRODUCTION

Let $A := (a_{ij})$ denote an n -by- n positive definite matrix. Consider a collection of index sets $\alpha_i \subseteq \{1, \dots, n\}$, $i = 1, \dots, p$ and define $A[\alpha_i]$ to be the principal submatrix of A at the intersection of the rows and columns indexed by α_i . By convention, $A[\emptyset] := 1$. Let $\alpha := \{\alpha_1, \dots, \alpha_p\}$ and define the product of determinants

$$A(\alpha) := \prod_{i=1}^p \det A[\alpha_i].$$

Much effort has been made to characterize the collection of index sets α and β for which the ratio $A(\alpha)/A(\beta)$ is bounded (more specifically, bounded by one) for all positive definite matrices A . Examples of these types of ratios are Hadamard's inequality and the more general Fisher's inequality. They are further generalized by

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Koteljanskii's inequality

$$\frac{\det A[\alpha_1 \cup \alpha_2] \det A[\alpha_1 \cap \alpha_2]}{\det A[\alpha_1] \det A[\alpha_2]} \leq 1, \quad (1)$$

which holds for any $\alpha_1, \alpha_2 \subseteq \{1, \dots, n\}$ and any positive definite A . For more information see Theorems 7.8.1, 7.8.5, and 7.8.9 in [11]. Various generalizations and extensions of inequality (1) appear in [1], [2], [3], [4], [6], [7], [9], [13], [14], [12], and many other works.

Finding other bounded determinantal ratios is a very difficult problem and proving them is even harder. Necessary and sufficient conditions (but not both) were formulated in [14]. In [10], the authors conjectured that

$$R_1(A) := \frac{A(\{\{1, 2, 4\}, \{1, 3, 4\}, \{2, 3\}, \{1\}, \{4\}\})}{A(\{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 4\}, \{3, 4\}\})} \leq \frac{27}{16}. \quad (2)$$

This upper bound was proven in [17], where a sequence of positive semidefinite matrices was given for which the ratio approaches 27/16.

Two more conjectures were stated in [10], namely that the supremums of the following ratios are 1

$$\begin{aligned} R_2(A) &= \frac{A(\{\{1, 2, 3, 4\}, \{1, 2, 3, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \{1\}, \emptyset\})}{A(\{\{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, \{2\}, \{3\}, \{4\}\})}, \\ R_3(A) &= \frac{A(\{\{1, 2, 3, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \{1\}, \{1\}, \emptyset\})}{A(\{\{2, 3, 4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2\}, \{3\}, \{4\}\})}, \end{aligned} \quad (3)$$

over all 4×4 positive definite matrices A .

By dividing appropriate rows and columns in the numerators and the denominators by the square root of the positive entries $a_{11}, a_{22}, a_{33}, a_{44}$, one can assume that A is a correlation matrix, that is

$$a_{11} = a_{22} = a_{33} = a_{44} = 1. \quad (4)$$

The positive definiteness implies that $a_{ij} \in (-1, 1)$ for all $1 \leq i < j \leq 4$. By multiplying rows and corresponding columns in the numerator and the denominator by -1 , if necessary, we can further assume that

$$a_{12}, a_{13}, a_{14} \in [0, 1). \quad (5)$$

An important observation is that both ratios $R_2(A)$ and $R_3(A)$ are symmetric with respect to the indices $\{2, 3, 4\}$. That is, the value of the ratio is invariant under any permutation of the set $\{2, 3, 4\}$.

To simplify the presentation, we adopt the notation $\Delta_\alpha := \det A[\alpha]$ for any index set $\alpha \subseteq \{1, 2, 3, 4\}$. For specific index sets, we simply list the indices as subscripts, e.g., $\Delta_{ij} := \det A[\{i, j\}]$. Under assumption (4), the ratios take the following form:

$$R_2(A) = \frac{\Delta_{1234}\Delta_{1234}\Delta_{23}\Delta_{24}\Delta_{34}}{\Delta_{123}\Delta_{124}\Delta_{134}\Delta_{234}} \quad \text{and} \quad R_3(A) = \frac{\Delta_{1234}\Delta_{23}\Delta_{24}\Delta_{34}}{\Delta_{234}\Delta_{12}\Delta_{13}\Delta_{14}}. \quad (6)$$

Our strategy is to look for the supremums of the ratios using optimization techniques. The domain over which we are optimizing is

$$D := \{A \in \mathbb{R}^{4 \times 4} : A \succ 0, \text{diag}(A) = (1, 1, 1, 1)\}. \quad (7)$$

In the remainder of the paper, we prove the following result.

Theorem 1.1. *We have $R_2(A) \leq 1$ and $R_3(A) \leq 1$ for all 4×4 positive definite matrices A .*

Proof. Suppose on the contrary that $\sup_D R_i > 1$, $i = 2, 3$. Choose a sequence $\{A_n\} \subset D$, such that $R_i(A_n)$ converges to $\sup_D R_i$. Since the closure of D in the affine space $\{A \in \mathbb{R}^{4 \times 4} : \text{diag}(A) = (1, 1, 1, 1)\}$ is compact, we can assume without loss of generality that $\{A_n\}$ converges to some positive (semi) definite correlation matrix A_0 . If $A_0 \in D$, then it has to be a critical point and $R_i(A_0) = 1$ as shown by Theorem 2.1. (Section 2 is devoted to the proof of that theorem.) Otherwise $A_0 \in \partial D$ and then $\limsup_{n \rightarrow \infty} R_i(A_n) \leq 1$, as shown in Section 3. These contradictions conclude the proof. $\square$

To avoid cumbersome indexing notation, we index the entries of any principal submatrix $A[\alpha]$ and its inverse using the original global indices from the full matrix A . Thus, the (i, j) entry of $A[\alpha]^{-1}$ corresponds to the original row i and column j of A , for $i, j \in \alpha$. For example, if $\alpha = \{2, 3\}$, the top-right entry of the 2×2 inverse matrix $A[\alpha]^{-1}$ is denoted as $(A[\alpha]^{-1})_{23}$ instead of the conventional $(A[\alpha]^{-1})_{12}$.

The following result, see [16, Theorem 6.1], is useful in the sequel.

Theorem 1.2. *Let M be a non-singular matrix partitioned as*

$$M = \begin{pmatrix} P & Q \\ R & S \end{pmatrix} \text{ and let } M^{-1} = \begin{pmatrix} \tilde{P} & \tilde{Q} \\ \tilde{R} & \tilde{S} \end{pmatrix} \quad (8)$$

be its inverse partitioned conformably. If the block $\tilde{S}$ is non-singular, then the principal submatrix P is also non-singular, and its inverse is given by the Schur's complement of $\tilde{S}$ in M^{-1} , namely

$$P^{-1} = \tilde{P} - \tilde{Q}\tilde{S}^{-1}\tilde{R}. \quad (9)$$

2. CRITICAL POINTS IN THE RELATIVE INTERIOR OF THE DOMAIN

The goal of this section is to prove the following theorem.

Theorem 2.1. *The critical points of the determinantal ratios $R_2(A)$ and $R_3(A)$, in the relative interior of D , satisfy*

$$\begin{aligned} a_{14} = a_{24} = a_{34} = 0 \text{ and } a_{23} = a_{12}a_{13}, \text{ or} \\ a_{12} = a_{24} = a_{23} = 0 \text{ and } a_{34} = a_{13}a_{14}, \text{ or} \\ a_{13} = a_{23} = a_{34} = 0 \text{ and } a_{24} = a_{12}a_{14}. \end{aligned}$$

Both ratios evaluate to 1 on the critical points.

2.1. The critical points of R_2 . We start by finding the system of equations for the critical points of R_2 in the relative interior of the domain. For that we analyze the partial derivatives of the logarithm of the original ratio $R_2(A)$

$$\begin{aligned} \ln R_2(A) = 2 \ln \det A + \ln \Delta_{23} + \ln \Delta_{24} + \ln \Delta_{34} \\ - \ln \Delta_{123} - \ln \Delta_{124} - \ln \Delta_{134} - \ln \Delta_{234}. \end{aligned}$$

Let $B = A^{-1}$ denote the inverse of the positive definite correlation matrix A . Note that

$$b_{ii} > 0, \text{ for all } i = 1, 2, 3, 4.$$

By Jacobi's formula, the derivative of the log-determinant of a positive definite matrix M with respect to an off-diagonal entry m_{ij} is given by

$$\frac{\partial \ln \det M}{\partial m_{ij}} = 2M_{ij}^{-1}.$$

Thus, taking the partial derivative of $\ln R_2(A)$ with respect to a_{12} , we observe that the terms $\Delta_{23}, \Delta_{24}, \Delta_{34}, \Delta_{134}$, and Δ_{234} vanish as they do not depend on a_{12} . This yields:

$$\frac{\partial \ln R_2(A)}{\partial a_{12}} = 4b_{12} - 2A[\{1, 2, 3\}]_{12}^{-1} - 2A[\{1, 2, 4\}]_{12}^{-1}. \quad (10)$$

To simplify the inverse submatrix entries, we utilize Theorem 1.2. Consider the partition of A and its inverse B obtained by isolating the fourth row and column:

$$A = \begin{pmatrix} A[\{1, 2, 3\}] & v \\ v^T & 1 \end{pmatrix}, \quad B = \begin{pmatrix} B[\{1, 2, 3\}] & u \\ u^T & b_{44} \end{pmatrix},$$

where $v = (a_{14}, a_{24}, a_{34})^T$ and $u = (b_{14}, b_{24}, b_{34})^T$. The inverse of the principal submatrix $A[\{1, 2, 3\}]$ can be expressed as the Schur's complement of b_{44} in B :

$$A[\{1, 2, 3\}]^{-1} = B[\{1, 2, 3\}] - \frac{uu^T}{b_{44}}.$$

Extracting the $(1, 2)$ entry from both sides of this matrix equation yields:

$$A[\{1, 2, 3\}]_{12}^{-1} = b_{12} - \frac{b_{14}b_{24}}{b_{44}}.$$

By applying identical partition logic to isolate the third row and column, we obtain the analogous expression for the submatrix on indices $\{1, 2, 4\}$:

$$A[\{1, 2, 4\}]_{12}^{-1} = b_{12} - \frac{b_{13}b_{23}}{b_{33}}.$$

Substituting these identities in (10) cancels the b_{12} terms, leaving:

$$\frac{\partial \ln R_2(A)}{\partial a_{12}} = 2 \left(\frac{b_{14}b_{24}}{b_{44}} + \frac{b_{13}b_{23}}{b_{33}} \right).$$

By the permutation symmetry of the indices $\{2, 3, 4\}$, differentiating with respect to a_{13} and a_{14} yields two analogous equations. Clearing the positive denominators, we obtain the first three equations for the critical points:

$$b_{14}b_{24}b_{33} + b_{13}b_{23}b_{44} = 0, \quad (11)$$

$$b_{14}b_{34}b_{22} + b_{12}b_{23}b_{44} = 0, \quad (12)$$

$$b_{12}b_{24}b_{33} + b_{13}b_{34}b_{22} = 0. \quad (13)$$

Next, we consider the partial derivative with respect to a_{23} :

$$\frac{\partial \ln R_2(A)}{\partial a_{23}} = 4b_{23} + 2A[\{2, 3\}]_{23}^{-1} - 2A[\{1, 2, 3\}]_{23}^{-1} - 2A[\{2, 3, 4\}]_{23}^{-1}. \quad (14)$$

Applying Theorem 1.2 again to the 3×3 submatrices gives:

$$A[\{1, 2, 3\}]_{23}^{-1} = b_{23} - \frac{b_{24}b_{34}}{b_{44}} \quad \text{and} \quad A[\{2, 3, 4\}]_{23}^{-1} = b_{23} - \frac{b_{12}b_{13}}{b_{11}}.$$

Substituting these into (14) cancels the $4b_{23}$ terms. Noting that the off-diagonal entry of the 2×2 inverse matrix is $A[\{2, 3\}]_{23}^{-1} = -a_{23}/(1 - a_{23}^2)$, we obtain:

$$\frac{\partial \ln R_2(A)}{\partial a_{23}} = 2 \left(\frac{-a_{23}}{1 - a_{23}^2} + \frac{b_{24}b_{34}}{b_{44}} + \frac{b_{12}b_{13}}{b_{11}} \right).$$

By symmetry, differentiating with respect to a_{24} and a_{34} provides the remaining two equations, completing the full system of six conditions for the critical points:

$$\frac{a_{23}}{1 - a_{23}^2} = \frac{b_{24}b_{34}}{b_{44}} + \frac{b_{12}b_{13}}{b_{11}}, \quad (15)$$

$$\frac{a_{24}}{1 - a_{24}^2} = \frac{b_{23}b_{34}}{b_{33}} + \frac{b_{12}b_{14}}{b_{11}}, \quad (16)$$

$$\frac{a_{34}}{1 - a_{34}^2} = \frac{b_{23}b_{24}}{b_{22}} + \frac{b_{13}b_{14}}{b_{11}}. \quad (17)$$

We now proceed to solving the system (11)–(13), (15)–(17). Rewrite (13) as $b_{12}b_{24}b_{33} = -b_{13}b_{34}b_{22}$. Multiply equation (11) by b_{12} and substitute this relation to get:

$$b_{14}(-b_{13}b_{34}b_{22}) + b_{13}(b_{12}b_{23}b_{44}) = 0. \quad (18)$$

From equation (12), we know $b_{12}b_{23}b_{44} = -b_{14}b_{34}b_{22}$. Substituting this into the second term gives $-2b_{13}b_{14}b_{34}b_{22} = 0$. Since $b_{22} > 0$, we obtain $b_{13}b_{14}b_{34} = 0$. The permutation symmetry with respect of the indices $\{2, 3, 4\}$, yields two additional constraints: $b_{12}b_{14}b_{24} = 0$ and $b_{12}b_{13}b_{23} = 0$.

Case 1. Suppose all entries b_{12}, b_{13}, b_{14} are non-zero. This forces $b_{23} = b_{24} = b_{34} = 0$. Consequently, the off-diagonal entries of B are non-zero only along the first row and column. The identity $AB = I_4$ dictates that the off-diagonal entries of the product must vanish. Specifically, for $j \in \{2, 3, 4\}$, the condition $(AB)_{1j} = 0$ together with $a_{11} = 1$ yields

$$b_{1j} = -a_{1j}b_{jj}, \text{ for } j \in \{2, 3, 4\}. \quad (19)$$

Furthermore, for distinct $i, j \in \{2, 3, 4\}$, the condition $(AB)_{ij} = 0$ expands to $a_{i1}b_{1j} + a_{ij}b_{jj} = 0$. Substituting b_{1j} into this equation gives $-a_{i1}a_{1j}b_{jj} + a_{ij}b_{jj} = 0$. Dividing by b_{jj} implies that the entries of A must satisfy the relations

$$a_{ij} = a_{i1}a_{1j}, \text{ for all } 2 \leq i < j \leq 4.$$

Next, for $j \in \{2, 3, 4\}$, the condition $(AB)_{jj} = 1$ expands to $a_{j1}b_{1j} + a_{jj}b_{jj} = 1$. Substituting $b_{1j} = -a_{1j}b_{jj}$ and $a_{jj} = 1$, we obtain $-a_{1j}^2b_{jj} + b_{jj} = 1$, which yields:

$$b_{jj} = \frac{1}{1 - a_{1j}^2}, \text{ for } j \in \{2, 3, 4\}. \quad (20)$$

Similarly, evaluating $(AB)_{11} = 1$ gives $b_{11} + a_{12}b_{12} + a_{13}b_{13} + a_{14}b_{14} = 1$. Substituting b_{1j} and the expressions for b_{jj} from (20), we derive a representation for b_{11} :

$$b_{11} = 1 + \frac{a_{12}^2}{1 - a_{12}^2} + \frac{a_{13}^2}{1 - a_{13}^2} + \frac{a_{14}^2}{1 - a_{14}^2}. \quad (21)$$

Now, we substitute the established relations into the partial derivative equations. Consider equation (15). Because $b_{24} = b_{34} = 0$, the first term on the right-hand side vanishes. Using (19) and $a_{23} = a_{12}a_{13}$, transforms the equation into:

$$\frac{a_{12}a_{13}}{1 - a_{12}^2a_{13}^2} = \frac{(-a_{12}b_{22})(-a_{13}b_{33})}{b_{11}} = \frac{a_{12}a_{13}b_{22}b_{33}}{b_{11}}. \quad (22)$$

Under the assumption in this case $b_{1j} \neq 0$, so the elements a_{12} and a_{13} are non-zero. Canceling $a_{12}a_{13}$ from both sides, solving for b_{11} , and substituting b_{22} and b_{33} from

(20), gives a second representation for b_{11} :

$$b_{11} = \frac{1 - a_{12}^2 a_{13}^2}{(1 - a_{12}^2)(1 - a_{13}^2)} = 1 + \frac{a_{12}^2}{1 - a_{12}^2} + \frac{a_{13}^2}{1 - a_{13}^2}. \quad (23)$$

Equating the two expressions for b_{11} from (21) and (23) forces $a_{14}^2/(1 - a_{14}^2) = 0$, which dictates $a_{14} = 0$. By (19), we have $b_{14} = 0$, which contradicts the assumption in this case, showing that this case is not possible.

Case 2. Suppose at least one of the entries b_{12}, b_{13}, b_{14} is zero. Without loss of generality, due to index symmetry, assume $b_{14} = 0$. Substituting $b_{14} = 0$ into equations (11)–(13), reduces them to:

$$\begin{aligned} b_{13}b_{23}b_{44} &= 0, \\ b_{12}b_{23}b_{44} &= 0, \\ b_{12}b_{24}b_{33} + b_{13}b_{34}b_{22} &= 0. \end{aligned} \quad (24)$$

Since the diagonal entries b_{44} and b_{22} are strictly positive, the first two equations imply that either $b_{23} = 0$, or $b_{12} = b_{13} = 0$.

Case 2a. If $b_{23} = 0$, we substitute $b_{14} = 0$ and $b_{23} = 0$ into equations (16) and (17). The right-hand sides vanish completely, yielding $a_{24} = 0$ and $a_{34} = 0$. With $a_{24} = a_{34} = 0$, the entry b_{14} of the inverse matrix evaluates strictly to $-a_{14}(1 - a_{23}^2)/\det A$. Since A is positive definite, the principal minor satisfies $(1 - a_{23}^2) > 0$. Thus, the constraint $b_{14} = 0$ implies $a_{14} = 0$. In a similar way, the constraint $b_{23} = 0$ implies the condition $a_{23} = a_{12}a_{13}$. Let us see now that equation (15) holds. Indeed, the conditions $a_{14} = a_{24} = a_{34} = 0$ imply that $b_{24} = b_{34} = 0$ and $b_{44} = 1$. This verifies equation (24). With $a_{23} = a_{12}a_{13}$, the remaining entries of the inverse matrix B are:

$$b_{11} = \frac{1 - a_{12}^2 a_{13}^2}{(1 - a_{12}^2)(1 - a_{13}^2)}, \quad b_{12} = \frac{-a_{12}}{1 - a_{12}^2}, \quad b_{13} = \frac{-a_{13}}{1 - a_{13}^2}. \quad (25)$$

Substituting these into the right-hand side of (15) verifies its validity. We are free to choose any values of a_{12} and a_{13} that maintain positive definiteness. Thus, this case gives the first solution in Theorem 2.1. The rest are derived by the symmetry.

Case 2b. If $b_{12} = b_{13} = 0$, then together with $b_{14} = 0$, the first row and column of the inverse matrix B contain zeros off the diagonal. This implies that $a_{12} = a_{13} = a_{14} = 0$. Equations (11)–(13) are automatically satisfied. Under these conditions, the entries of B in terms of those of A are $b_{11} = 1$ and

$$\begin{aligned} b_{22} &= \frac{1 - a_{34}^2}{\det A}, \quad b_{23} = -\frac{a_{23} - a_{24}a_{34}}{\det A}, \quad b_{24} = -\frac{a_{24} - a_{23}a_{34}}{\det A}, \\ b_{33} &= \frac{1 - a_{24}^2}{\det A}, \quad b_{34} = -\frac{a_{34} - a_{23}a_{24}}{\det A}, \quad b_{44} = \frac{1 - a_{23}^2}{\det A}. \end{aligned}$$

Substituting into equations (15)–(17) and simplifying gives

$$a_{24}a_{34} = a_{23}, \quad a_{23}a_{34} = a_{24}, \quad a_{23}a_{24} = a_{34}.$$

The only solution of this system that gives a positive definite A is $a_{23} = a_{24} = a_{34} = 0$. This is a special case of the first solution in Theorem 2.1.

It is straightforward to verify that when $a_{14} = a_{24} = a_{34} = 0$ and $a_{23} = a_{12}a_{13}$, we have $R_2(A) = 1$. Consequently, to establish the global inequality $R_2(A) \leq 1$, it remains to verify that the ratio does not strictly exceed 1 as the matrix approaches the boundary of the domain.

2.2. The critical points of R_3 . We begin by characterizing the critical points of R_3 in the relative interior of its domain. We look at the partial derivatives of the logarithm of $R_3(A)$:

$$\ln R_3(A) = \ln \det A + \ln \Delta_{23} + \ln \Delta_{24} + \ln \Delta_{34} - \ln \Delta_{234} - \ln \Delta_{12} - \ln \Delta_{13} - \ln \Delta_{14}.$$

Taking the partial derivative with respect to a_{12} , we observe that the terms involving $\Delta_{23}, \Delta_{24}, \Delta_{34}, \Delta_{234}, \Delta_{13}$, and Δ_{14} vanish. This yields:

$$\frac{\partial \ln R_3(A)}{\partial a_{12}} = 2b_{12} - 2A[\{1, 2\}]_{12}^{-1}.$$

For the 2×2 principal submatrix $A[\{1, 2\}]$, its inverse can be explicitly computed. Extracting the off-diagonal entry gives $A[\{1, 2\}]_{12}^{-1} = -a_{12}/(1 - a_{12}^2)$. Substituting this into the derivative equation yields:

$$b_{12} + \frac{a_{12}}{1 - a_{12}^2} = 0. \quad (26)$$

By the permutation symmetry of the indices $\{2, 3, 4\}$, differentiating with respect to a_{13} and a_{14} yields two analogous equations:

$$b_{13} + \frac{a_{13}}{1 - a_{13}^2} = 0, \quad (27)$$

$$b_{14} + \frac{a_{14}}{1 - a_{14}^2} = 0. \quad (28)$$

Next, we consider the partial derivative with respect to a_{23} :

$$\frac{\partial \ln R_3(A)}{\partial a_{23}} = 2b_{23} + 2A[\{2, 3\}]_{23}^{-1} - 2A[\{2, 3, 4\}]_{23}^{-1}.$$

To simplify the inverse submatrix entries, consider the partition of A and its inverse $B = A^{-1}$ isolating the first row and column:

$$A = \begin{pmatrix} 1 & v^T \\ v & A[\{2, 3, 4\}] \end{pmatrix}, \quad B = \begin{pmatrix} b_{11} & u^T \\ u & B[\{2, 3, 4\}] \end{pmatrix},$$

where $v = (a_{12}, a_{13}, a_{14})^T$ and $u = (b_{12}, b_{13}, b_{14})^T$. By Theorem 1.2, the inverse of the principal submatrix $A[\{2, 3, 4\}]$ is the Schur's complement of b_{11} in B :

$$A[\{2, 3, 4\}]^{-1} = B[\{2, 3, 4\}] - \frac{uu^T}{b_{11}}.$$

Extracting the entry corresponding to the indices $\{2, 3\}$ from both sides of this matrix equation gives:

$$A[\{2, 3, 4\}]_{23}^{-1} = b_{23} - \frac{b_{12}b_{13}}{b_{11}}.$$

Additionally, the off-diagonal entry of the 2×2 inverse matrix is $A[\{2, 3\}]_{23}^{-1} = -a_{23}/(1 - a_{23}^2)$. Substituting these two identities into the derivative cancels the $2b_{23}$ terms and after easy simplification, we obtain

$$\frac{b_{12}b_{13}}{b_{11}} - \frac{a_{23}}{1 - a_{23}^2} = 0. \quad (29)$$

By symmetry, differentiating with respect to a_{24} and a_{34} provides the remaining two equations, completing the full system of six equations:

$$\frac{b_{12}b_{14}}{b_{11}} - \frac{a_{24}}{1 - a_{24}^2} = 0, \quad (30)$$

$$\frac{b_{13}b_{14}}{b_{11}} - \frac{a_{34}}{1 - a_{34}^2} = 0. \quad (31)$$

To solve this system, we look at the matrix identity $AB = I_4$. For brevity, we introduce the auxiliary variables $x_j = a_{1j}/(1 - a_{1j}^2)$ for $j \in \{2, 3, 4\}$. From the first subsystem (26)–(28), the off-diagonal entries in the first row of B can be expressed as $b_{1j} = -x_j$.

Evaluating the $(1, 1)$ entry of the product $AB = I_4$ yields

$$a_{11}b_{11} + a_{12}b_{12} + a_{13}b_{13} + a_{14}b_{14} = 1.$$

Substituting $a_{11} = 1$ and $b_{1j} = -x_j$ provides an explicit representation for b_{11} :

$$b_{11} = 1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4. \quad (32)$$

Next, the $(2, 1)$ entry of $AB = I_4$ is

$$a_{12}b_{11} + a_{22}b_{12} + a_{23}b_{13} + a_{24}b_{14} = 0.$$

Substitute $a_{22} = 1$ and $b_{j1} = -x_j$ to obtain:

$$a_{12}b_{11} - x_2 - a_{23}x_3 - a_{24}x_4 = 0.$$

Substituting the explicit formula for b_{11} from (32) into this equation gives:

$$\begin{aligned} & a_{12}(1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4) - x_2 - a_{23}x_3 - a_{24}x_4 \\ &= a_{12} - (1 - a_{12}^2)x_2 + (a_{12}a_{13} - a_{23})x_3 + (a_{12}a_{14} - a_{24})x_4 \\ &= (a_{12}a_{13} - a_{23})x_3 + (a_{12}a_{14} - a_{24})x_4 = 0, \end{aligned}$$

where we used the definition of x_2 . By the permutation symmetry of the indices $\{2, 3, 4\}$, evaluating the $(3, 1)$ and $(4, 1)$ entries of $AB = I_4$ generates two analogous equations. Defining $U := a_{12}a_{13} - a_{23}$, $V := a_{12}a_{14} - a_{24}$, and $W := a_{13}a_{14} - a_{34}$, we obtain a homogeneous linear system in terms of the ‘variables’ (U, V, W) :

$$\begin{aligned} Ux_3 + Vx_4 &= 0, \\ Ux_2 + Wx_4 &= 0, \\ Vx_2 + Wx_3 &= 0. \end{aligned} \quad (33)$$

The determinant of this system is $-2x_2x_3x_4$.

Case 1. Suppose the entries a_{12}, a_{13}, a_{14} are all non-zero. Consequently, x_2, x_3, x_4 are non-zero, making the determinant strictly non-zero. This forces the homogeneous system to have only the trivial solution: $U = V = W = 0$. In particular, $U = 0$ implies $a_{23} = a_{12}a_{13}$.

However, substituting $a_{23} = a_{12}a_{13}$ back into the partial derivative equation (29) yields:

$$\frac{a_{12}a_{13}}{(1 - a_{12}^2)(1 - a_{13}^2)b_{11}} - \frac{a_{12}a_{13}}{1 - a_{12}^2a_{13}^2} = 0.$$

Solving this for b_{11} gives:

$$b_{11} = \frac{1 - a_{12}^2a_{13}^2}{(1 - a_{12}^2)(1 - a_{13}^2)} = 1 + \frac{a_{12}^2}{1 - a_{12}^2} + \frac{a_{13}^2}{1 - a_{13}^2}. \quad (34)$$

Comparing this with (32), one sees that the equality holds if and only if $a_{14}x_4 = a_{14}^2/(1 - a_{14}^2) = 0$, which mandates $a_{14} = 0$. This contradiction shows that this case is not possible.

Case 2. Suppose at least one of the entries a_{12}, a_{13}, a_{14} is zero. Without loss of generality, due to the symmetry, assume $a_{14} = 0$, which implies $x_4 = 0$ and $b_{14} = 0$.

Substituting $b_{14} = 0$ into the partial derivative equations (30) and (31) yields $a_{24}/(1 - a_{24}^2) = 0$ and $a_{34}/(1 - a_{34}^2) = 0$ implying that $a_{24} = 0$ and $a_{34} = 0$. Next, substituting $x_4 = 0$ into the homogeneous system (33) reduces the first equation to $Ux_3 = 0$.

Case 2a. If both a_{12} and a_{13} are not zero, then $x_3 \neq 0$, which forces $U = 0$, that is $a_{23} = a_{12}a_{13}$.

Case 2b. If $a_{12} = 0$, then $x_2 = 0$ and consequently $b_{12} = 0$. Substituting $b_{12} = 0$ into equation (29) eliminates the first term, leaving $-a_{23}/(1 - a_{23}^2) = 0$, which forces $a_{23} = 0$. The relationship $a_{23} = a_{12}a_{13}$ remains valid.

If $a_{13} = 0$, then we analogously conclude that $a_{23} = 0$ and the relationship $a_{23} = a_{12}a_{13}$ remains valid.

This gives the first solution in Theorem 2.1. The rest are derived by the symmetry.

It is straightforward to verify that when $a_{14} = a_{24} = a_{34} = 0$ and $a_{23} = a_{12}a_{13}$, we have $R_3(A) = 1$. Consequently, to establish the global inequality $R_3(A) \leq 1$, it remains to verify that the ratio does not exceed 1 as the matrix approaches the boundary of the domain.

3. BOUNDARY ANALYSIS

To prove the global inequalities $R_2(A) \leq 1$ and $R_3(A) \leq 1$, we need to analyze the behavior of the ratios as the correlation matrix A approaches the boundary of the positive definite cone. Throughout this section, we assume that

$$A \rightarrow A_0 \in \partial D,$$

where D is defined by (7). Since A satisfies assumption (5), the limit A_0 satisfies it too. Let $\lambda_1 \geq \dots \geq \lambda_4 > 0$ be the eigenvalues of A with corresponding orthonormal eigenvectors $u_1, \dots, u_4$. As $A \rightarrow A_0$, the eigenvalues λ_i converge to the eigenvalues μ_i of A_0 , and the eigenvectors $u_i := (u_{1i}, u_{2i}, u_{3i}, u_{4i})^T$ can be chosen to converge to corresponding orthonormal eigenvectors $v_i := (v_{1i}, v_{2i}, v_{3i}, v_{4i})^T$ of A_0 , $i = 1, \dots, 4$.

The inverse of A is given by

$$A^{-1} = \frac{1}{\lambda_4} u_4 u_4^T + \sum_{i=1}^3 \frac{1}{\lambda_i} u_i u_i^T =: \frac{1}{\lambda_4} u_4 u_4^T + M.$$

By the identity $\text{adj}(A) = (\det A)A^{-1}$, we have $\text{adj}(A) = \lambda_1 \lambda_2 \lambda_3 (u_4 u_4^T + \lambda_4 M)$. Evaluating the diagonal entries of this matrix, which correspond to the 3×3 principal minors of A , gives:

$$\begin{aligned} \Delta_{234}(A) &= \lambda_1 \lambda_2 \lambda_3 (u_{14}^2 + \lambda_4 M_{11}), & \Delta_{134}(A) &= \lambda_1 \lambda_2 \lambda_3 (u_{24}^2 + \lambda_4 M_{22}), \\ \Delta_{124}(A) &= \lambda_1 \lambda_2 \lambda_3 (u_{34}^2 + \lambda_4 M_{33}), & \Delta_{123}(A) &= \lambda_1 \lambda_2 \lambda_3 (u_{44}^2 + \lambda_4 M_{44}). \end{aligned} \quad (35)$$

3.1. Suppose A_0 has rank 3. In this case, the eigenvalues of A_0 are $\mu_1 \geq \mu_2 \geq \mu_3 > \mu_4 = 0$ with $A_0 v_4 = 0$ and $\|v_4\| = 1$.

3.1.1. Boundary behaviour of R_2 . Substituting (35) into $R_2(A)$ provides a path-independent representation of the ratio:

$$R_2(A) = \frac{\lambda_4^2 \Delta_{23}(A) \Delta_{24}(A) \Delta_{34}(A)}{(\lambda_1 \lambda_2 \lambda_3)^2 \prod_{i=1}^4 (u_{i4}^2 + \lambda_4 M_{ii})}. \quad (36)$$

We investigate the limit of this ratio based on the sparsity of the null vector v_4 . Note that v_4 cannot have three zero components. Indeed, if $v_{i4} \neq 0$, then the i -th row of $A_0 v_4 = 0$ evaluates to $v_{i4} = 0$, a contradiction.

Case 1. *Vector v_4 has no zero components.* In this case, the denominator of $R_2(A)$ approaches $(\mu_1\mu_2\mu_3)^2 v_{14}^2 \cdots v_{44}^2 > 0$. Since $\lambda_4^2 \rightarrow 0$, we conclude that $\lim_{A \rightarrow A_0} R_2(A) = 0$.

Case 2. *The vector v_4 has exactly one zero component.* Suppose $v_{k4} = 0$ and $v_{i4} \neq 0$ for $i \neq k$. By definition,

$$M_{kk} = \sum_{r=1}^3 \frac{u_{kr}^2}{\lambda_r} \rightarrow \sum_{r=1}^3 \frac{v_{kr}^2}{\mu_r} =: 2c > 0,$$

since v_1, v_2, v_3, v_4 form an orthonormal basis and $v_{k4} = 0$ implies $\sum_{r=1}^3 v_{kr}^2 = 1$. Thus, for A sufficiently close to A_0 , we have

$$u_{k4}^2 + \lambda_4 M_{kk} \geq c\lambda_4.$$

The other factors $u_{i4}^2 + \lambda_4 M_{ii}$ converge to $v_{i4}^2 > 0$, $i \neq k$. Hence, by taking $c > 0$ smaller, if necessary, we have that for A sufficiently close to A_0 :

$$(\lambda_1 \lambda_2 \lambda_3)^2 \prod_{i=1}^4 (u_{i4}^2 + \lambda_4 M_{ii}) \geq c\lambda_4.$$

Since every 2×2 principal minor of a correlation matrix is at most 1, using (36), we get

$$0 \leq R_2(A) \leq \frac{\lambda_4^2}{c\lambda_4} = \frac{\lambda_4}{c} \rightarrow 0.$$

Case 3a. *Vector v_4 has exactly two zero components, both from the index set $\{2, 3, 4\}$.* Without loss of generality, assume $v_{34} = v_{44} = 0$ and $v_{14}, v_{24} \neq 0$. The condition $A_0 v_4 = 0$ applied to the first two rows yields $v_{14} + (A_0)_{12} v_{24} = 0$ and $(A_0)_{12} v_{14} + v_{24} = 0$. Since $(A_0)_{12} \geq 0$, the system implies that $(A_0)_{12} = 1$ and $v_{14} = -v_{24}$. Evaluating the third and fourth rows with $v_{14} = -v_{24}$ yields $(A_0)_{13} = (A_0)_{23}$ and $(A_0)_{14} = (A_0)_{24}$.

By Lemma 4.1, for the positive definite A we have

$$R_2(A) \leq \frac{\Delta_{23}(A) \Delta_{34}(A)}{\Delta_{13}(A)}.$$

Since $(A_0)_{13} = (A_0)_{23}$, we have $\Delta_{13}(A_0) = \Delta_{23}(A_0)$. If $\Delta_{13}(A_0) > 0$, then

$$\limsup_{A \rightarrow A_0} R_2(A) \leq \frac{\Delta_{23}(A_0) \Delta_{34}(A_0)}{\Delta_{13}(A_0)} = \Delta_{34}(A_0) \leq 1.$$

If $\Delta_{13}(A_0) = 0$, then $(A_0)_{13} = (A_0)_{23} = 1$. Since A_0 is a correlation matrix, its Gram representation, shows that columns 1, 2, and 3 are identical. This contradicts the assumption that the rank of A_0 is 3.

Case 3b. *Vector v_4 has exactly two zero components, one of which has index 1.* Due to the permutation symmetry, assume $v_{14} = v_{24} = 0$ and $v_{34}, v_{44} \neq 0$. The condition $A_0 v_4 = 0$ applied to the last two rows yields $v_{34} + (A_0)_{34} v_{44} = 0$ and $(A_0)_{34} v_{34} + v_{44} = 0$. This implies that $(A_0)_{34} = \pm 1$ (hence $\Delta_{34}(A_0) = 0$) and $v_{34} = \mp v_{44}$.

Evaluating the first two rows with $v_{34} = \mp v_{44}$ yields $(A_0)_{13} = \pm (A_0)_{14}$ and $(A_0)_{23} = \pm (A_0)_{24}$. If $(A_0)_{13} = (A_0)_{14} = 0$, then $\Delta_{13}(A_0) = 1$, and taking limit superior in the inequality in Lemma 4.1, one sees that the right-hand side is bounded above by one. Thus we can suppose that $(A_0)_{13} = (A_0)_{14} > 0$ and one should choose the top sign everywhere above. That is, $(A_0)_{34} = 1$, $(A_0)_{23} = (A_0)_{24}$, and $v_{34} = -v_{44}$.

If $\Delta_{13}(A_0) > 0$, then taking limit superior in the inequality in Lemma 4.1, one sees that the right-hand side converges to zero, since $\Delta_{34}(A_0) = 0$.

If $\Delta_{13}(A_0) = 0$, then $(A_0)_{13} = 1$. Since A_0 is a correlation matrix, its Gram representation, together with $(A_0)_{34} = 1$, shows that x_1, x_3, x_4 are identical. This contradicts the assumption that the rank of A_0 is 3.

3.1.2. *Boundary behaviour of R_3 .* Using (35), the ratio $R_3(A)$ can be expressed path-independently as follows

$$R_3(A) = \frac{\lambda_4 \Delta_{23}(A) \Delta_{24}(A) \Delta_{34}(A)}{(u_{14}^2 + \lambda_4 M_{11}) \Delta_{12}(A) \Delta_{13}(A) \Delta_{14}(A)}.$$

We consider two cases based on the first component of the null vector v_4 .

Case 1. Suppose $v_{14} \neq 0$. The term $u_{14}^2 + \lambda_4 M_{11}$ converges to $v_{14}^2 > 0$. If the principal minors of A_0 satisfy $\Delta_{1j}(A_0) > 0$ for all $j \in \{2, 3, 4\}$, the denominator converges to a positive constant. Since $\lambda_4 \rightarrow 0$, we have $R_3(A) \rightarrow 0$.

If any of these minors vanish, then by symmetry of the indexes $\{2, 3, 4\}$, we can assume $\Delta_{12}(A_0) = 0$. Then $(A_0)_{12} = 1$ implies $x_1 = x_2$, and therefore $\Delta_{13}(A_0) = \Delta_{23}(A_0)$ and $\Delta_{14}(A_0) = \Delta_{24}(A_0)$. If $\Delta_{13}(A_0) > 0$, then by Lemma 4.1

$$\limsup_{A \rightarrow A_0} R_3(A) \leq \limsup_{A \rightarrow A_0} \frac{\Delta_{23}(A) \Delta_{34}(A)}{\Delta_{13}(A)} = \frac{\Delta_{23}(A_0) \Delta_{34}(A_0)}{\Delta_{13}(A_0)} = \Delta_{34}(A_0) \leq 1.$$

If $\Delta_{13}(A_0) = 0$, then $x_1 = x_2 = x_3$, so the first, second, and third columns of A_0 are identical, contradicting the assumption that the rank of A_0 is 3.

Case 2. Suppose $v_{14} = 0$. The condition $A_0 v_4 = 0$ implies that the principal submatrix $A_0[\{2, 3, 4\}]$ has a non-trivial null vector $(v_{24}, v_{34}, v_{44})^T$, yielding $\Delta_{234}(A_0) = 0$. Geometrically, this means the Gram vectors x_2, x_3, x_4 associated with A_0 span a subspace of dimension at most 2. Because the rank of A_0 is 3, the vector x_1 cannot lie in this subspace. Decompose x_1 as $x_1 = \ell y + x_1^\perp$, where ℓy is the orthogonal projection of x_1 onto the span of x_2, x_3, x_4 and $x_1^\perp := x_1 - \ell y$ is orthogonal to that span. We assume that $\|y\| = 1$ and $\ell \geq 0$. Since $\|x_1\| = 1$, we have $|\ell| \leq 1$.

If $\ell = 0$, then $x_1 \perp x_j$ for $j = 2, 3, 4$, so $\Delta_{12}(A_0) = \Delta_{13}(A_0) = \Delta_{14}(A_0) = 1$. Taking limit superior from both sides of the inequality in Lemma 4.1, one sees that the right-hand side is bounded above by one.

Assume now $\ell > 0$. By the sign assumption on the entries of A_0 , we have $x_1^T x_j \geq 0$, hence $y^T x_j \geq 0$, and

$$\Delta_{1j}(A_0) = 1 - \ell^2 (y^T x_j)^2 \geq 1 - (y^T x_j)^2 =: \Delta_{yj} \text{ for } j = 2, 3, 4. \quad (37)$$

The three numbers $\Delta_{y2}, \Delta_{y3}, \Delta_{y4}$ cannot all be zero, otherwise x_2, x_3, x_4 would all be parallel to y , so their span would be one-dimensional, and A_0 would have rank at most 2, a contradiction. Apply Lemma 4.2 to the correlation matrix, A_y , generated by y, x_2, x_3, x_4 and having rank at most 2, to obtain that

$$\min \left(\frac{\Delta_{23}(A_y) \Delta_{34}(A_y)}{\Delta_{y3}}, \frac{\Delta_{23}(A_y) \Delta_{24}(A_y)}{\Delta_{y2}}, \frac{\Delta_{24}(A_y) \Delta_{34}(A_y)}{\Delta_{y4}} \right) \leq 1, \quad (38)$$

where the minimum is taken over the well-defined ratios.

Suppose, without loss of generality, that the first ratio is well-defined and attains the minimum. Thus, $\Delta_{y3} > 0$ and $\Delta_{13}(A_0) \geq \Delta_{y3} > 0$. By Lemma 4.1,

$$\limsup_{A \rightarrow A_0} R_3(A) \leq \limsup_{A \rightarrow A_0} \frac{\Delta_{23}(A) \Delta_{34}(A)}{\Delta_{13}(A)} = \frac{\Delta_{23}(A_0) \Delta_{34}(A_0)}{\Delta_{13}(A_0)}$$

$$\leq \frac{\Delta_{23}(A_y)\Delta_{34}(A_y)}{\Delta_{y3}} \leq 1,$$

where we used the fact that $\Delta_{ij}(A_0) = \Delta_{ij}(A_y)$ for $ij = 23, 24, 34$.

3.2. Suppose A_0 has rank at most 2. If the minors $\Delta_{12}(A_0)$, $\Delta_{13}(A_0)$, $\Delta_{14}(A_0)$ are not all zero, then without loss of generality assume that $\Delta_{12}(A_0) > 0$ and the minimum in Lemma 4.2 is attained by its first ratio. Taking limit superior from both sides of the inequality in Lemma 4.1, shows that $\limsup_{A \rightarrow A_0} R_i(A) \leq 1$ for $i = 2, 3$.

If $\Delta_{12}(A_0) = \Delta_{13}(A_0) = \Delta_{14}(A_0) = 0$, then $(A_0)_{12} = (A_0)_{13} = (A_0)_{14} = 1$ implying that $x_1 = x_2 = x_3 = x_4$ and so A_0 is the all-one matrix. Let $A_n = (a_{ij}^n)$ be a sequence of positive definite correlation matrices approaching A_0 . Write

$$\delta_{ij}^n := 1 - a_{ij}^n, \quad \varepsilon_n := \max_{i < j} \delta_{ij}^n \quad \text{and} \quad m_n := \max_{j=2,3,4} \delta_{1j}^n.$$

For n large, we have $a_{ij}^n \in (0, 1)$ and $\varepsilon_n \rightarrow 0$. Let $x_1^n, \dots, x_4^n$ be unit vectors whose Gram matrix is A_n . Then

$$\|x_i^n - x_j^n\|^2 = 2(1 - a_{ij}^n) = 2\delta_{ij}^n.$$

We claim that $m_n \geq \varepsilon_n/4$. Indeed, if the maximum ε_n is attained by a pair of indexes involving index 1, this is immediate. Otherwise, suppose it is attained by a pair $i, j \in \{2, 3, 4\}$. Then by the triangle inequality, we have

$$\sqrt{2\varepsilon_n} = \|x_i^n - x_j^n\| \leq \|x_i^n - x_1^n\| + \|x_1^n - x_j^n\| \leq 2\sqrt{2m_n},$$

hence $m_n \geq \varepsilon_n/4$.

For each n , choose $j_n \in \{2, 3, 4\}$, such that $\delta_{1j_n}^n = m_n$. On the one hand, we have

$$\Delta_{1j_n}(A_n) = 1 - (a_{1j_n}^n)^2 = (1 - a_{1j_n}^n)(1 + a_{1j_n}^n) \geq \delta_{1j_n}^n = m_n \geq \varepsilon_n/4,$$

On the other hand, for every $k < \ell$, we have

$$\Delta_{k\ell}(A_n) = (1 - a_{k\ell}^n)(1 + a_{k\ell}^n) \leq 2\delta_{k\ell}^n \leq 2\varepsilon_n.$$

Using Lemma 4.1, choose the term whose denominator is $\Delta_{1j_n}(A_n)$. Its numerator is a product of two among $\Delta_{23}, \Delta_{24}, \Delta_{34}$. Therefore, for $i = 2, 3$, we have

$$R_i(A_n) \leq \frac{(2\varepsilon_n)^2}{\varepsilon_n/4} = 16\varepsilon_n \rightarrow 0.$$

4. APPENDIX

Lemma 4.1. *For every positive definite correlation matrix A , we have*

$$R_i(A) \leq \min \left(\frac{\Delta_{23}\Delta_{24}}{\Delta_{12}}, \frac{\Delta_{23}\Delta_{34}}{\Delta_{13}}, \frac{\Delta_{24}\Delta_{34}}{\Delta_{14}} \right)$$

for $i = 2, 3$.

Proof. Koteljanskii's inequality gives $\Delta_{1234}\Delta_{24} \leq \Delta_{124}\Delta_{234}$, and $\Delta_{1234}\Delta_{13} \leq \Delta_{123}\Delta_{134}$. Multiplying these inequalities and substituting into the definition of R_2 yields

$$R_2(A) \leq \frac{\Delta_{23}\Delta_{34}}{\Delta_{13}}.$$

The other two bounds follow by the symmetry in the indices 2, 3, 4.

Again by Koteljanskii $\Delta_{1234}\Delta_{23} \leq \Delta_{123}\Delta_{234}$. Substituting into R_3 , gives

$$R_3(A) \leq \frac{\Delta_{123}\Delta_{24}\Delta_{34}}{\Delta_{12}\Delta_{13}\Delta_{14}} \leq \frac{\Delta_{24}\Delta_{34}}{\Delta_{14}},$$

where the last inequality holds since A is a correlation matrix and by Koteljanskii $\Delta_{123} \leq \Delta_{12}\Delta_{13}$. Again, the remaining two bounds follow from the symmetry in 2, 3, 4. $\square$

When the correlation matrix has low rank, the minimum in Lemma 4.1 can be estimated directly.

Lemma 4.2. *Let A_0 be a 4×4 positive semidefinite correlation matrix with rank at most 2. Then either*

$$\Delta_{12} = \Delta_{13} = \Delta_{14} = 0,$$

or at least one of the following ratios is well-defined and

$$\min \left(\frac{\Delta_{23}\Delta_{24}}{\Delta_{12}}, \frac{\Delta_{23}\Delta_{34}}{\Delta_{13}}, \frac{\Delta_{24}\Delta_{34}}{\Delta_{14}} \right) \leq 1,$$

where the minimum is taken over the well-defined ratios.

Proof. Let A_0 be the Gram matrix of unit vectors $x_1, x_2, x_3, x_4 \in \mathbb{R}^2$. Since the quantities

$$\Delta_{ij} = 1 - (x_i^T x_j)^2$$

are unchanged if any x_j is replaced by $-x_j$, we may choose the signs of x_2, x_3, x_4 so that $x_1^T x_j \geq 0$, $j = 2, 3, 4$. This does not change any of the Δ_{ij} . Rotate the plane so that $x_1 = (1, 0)$. Then we may write

$$x_j = (\cos \theta_j, \sin \theta_j), \quad \theta_j \in [-\pi/2, \pi/2], \quad j = 2, 3, 4.$$

Hence $\Delta_{1j} = \sin^2 \theta_j$ and $\Delta_{ij} = \sin^2(\theta_i - \theta_j)$.

Order the three angles:

$$\alpha \leq \beta \leq \gamma,$$

and let the corresponding indices be p, q, r , so that

$$\theta_p = \alpha, \quad \theta_q = \beta, \quad \theta_r = \gamma.$$

If $\Delta_{12} = \Delta_{13} = \Delta_{14} = 0$, we are in the exceptional case. Otherwise, not all of α, β, γ are zero. Thus either $\alpha < 0$ or $0 < \gamma$. We consider two cases depending on the sign of β .

If $\gamma \leq 0$, then we must have $\alpha < 0$ and $\beta \leq 0$. Then $0 \leq \beta - \alpha \leq -\alpha \leq \pi/2$, so

$$|\sin(\beta - \alpha)| \leq |\sin \alpha|.$$

Since $|\sin(\gamma - \alpha)| \leq 1$ and $|\sin \alpha| > 0$, we conclude

$$\frac{\Delta_{pq}\Delta_{pr}}{\Delta_{1p}} = \left(\frac{|\sin(\beta - \alpha)| |\sin(\gamma - \alpha)|}{|\sin \alpha|} \right)^2 \leq 1.$$

If $0 \leq \alpha$, then we must have $0 \leq \beta$ and $0 < \gamma$. Then $0 \leq \gamma - \beta \leq \gamma \leq \pi/2$, so

$$|\sin(\gamma - \beta)| \leq \sin \gamma.$$

Since $|\sin(\gamma - \alpha)| \leq 1$ and $\sin \gamma > 0$, we have

$$\frac{\Delta_{pr}\Delta_{qr}}{\Delta_{1r}} = \left(\frac{|\sin(\gamma - \alpha)| |\sin(\gamma - \beta)|}{\sin \gamma} \right)^2 \leq 1.$$

Thus one of the three well-defined ratios is at most 1. $\square$

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